

Table 17: Application A1, Scenario 1: Autonomous math problem-solving. Preliminary test with CAMEL. We constructed two roles: “Math Solver” and “Python Programmer”. The resulting code cannot be executed. The conversation continues, and the roles are not keen on solving the problem and giving an answer.

	Action	CAMEL
	Set task prompt	Completely simplify and rationalize the denominator: $\frac{\sqrt{160}}{\sqrt{252}} \times \frac{\sqrt{245}}{\sqrt{108}}$
	System	*****Assistant sys message***** AI SystemMessage(role_name='Python Programmer', role_type=<RoleType.ASSISTANT: 'assistant'>, meta_dict='task': 'Using Python programming language, write a program that accepts four integers as input, applies square root function to each, simplifies and rationalizes the denominator of the fraction resulting from the multiplication of two fractions: $\frac{\sqrt{160}}{\sqrt{252}} \times \frac{\sqrt{245}}{\sqrt{108}}$, and returns the simplified expression.', 'assistant_role': 'Python Programmer', 'user_role': 'Math Solver', role='system', content='Never forget you are a Python Programmer and I am a Math Solver. Never flip roles! ...) *****AI User sys message***** SystemMessage(role_name='Math Solver', role_type=<RoleType.USER: 'user'>, meta_dict='task': 'Using Python programming language, write a program that accepts four integers as input, applies square root function to each, simplifies and rationalizes the denominator of the fraction resulting from the multiplication of two fractions: $\frac{\sqrt{160}}{\sqrt{252}} \times \frac{\sqrt{245}}{\sqrt{108}}$, and returns the simplified expression.', 'assistant_role': 'Python Programmer', 'user_role': 'Math Solver', role='system', content='Never forget you are a Math Solver and I am a Python Programmer. Never flip roles! You will always instruct me. We share a common interest in collaborating to successfully complete a task. I must help you to complete the task. Here is the task: Using Python programming language, write a program that accepts four integers as input, applies square root function to each, simplifies and rationalizes the denominator of the fraction resulting from the multiplication of two fractions: $\frac{\sqrt{160}}{\sqrt{252}} \times \frac{\sqrt{245}}{\sqrt{108}}$, and returns the simplified expression.. ...) *****Original task prompt***** Completely simplify and rationalize the denominator: $\frac{\sqrt{160}}{\sqrt{252}} \times \frac{\sqrt{245}}{\sqrt{108}}$ *****Specified task prompt***** Using Python programming language, write a program that accepts four integers as input, applies square root function to each, simplifies and rationalizes the denominator of the fraction resulting from the multiplication of two fractions: $\frac{\sqrt{160}}{\sqrt{252}} \times \frac{\sqrt{245}}{\sqrt{108}}$, and returns the simplified expression. *****Final task prompt***** Using Python programming language, write a program that accepts ...
	AI User	Instruction: Import the necessary libraries for the task. Input: None
	AI Assistant	Solution: To solve this task, we will need the 'math' library for the square root function and the 'fractions' library to simplify the resulting fraction. Here is how to import these libraries: <pre>1 import math 2 from fractions import Fraction</pre> The 'math' library provides functions that deal with mathematical operations. The 'sqrt' function, which we will use in this task, is part of this library. The 'fractions' library provides functions for creating and manipulating rational numbers. The 'Fraction' function, which we will use to simplify the resulting fraction, is part of this library. Next request.
	AI User	Instruction: Define a function that accepts four integers as input. Input: None
	AI Assistant	Solution: Let's define a function named 'simplify_fraction' that accepts four integers as input. These integers will represent the numbers under the square root in the fractions. <pre>1 def simplify_fraction(a, b, c, d): 2 pass</pre> In this function, 'a', 'b', 'c', and 'd' are the parameters that will accept the four integers as input. The 'pass' statement is a placeholder and will be replaced with actual code in the next steps. Next request.
	AI User	Instruction: Inside the function, apply the square root function to each of the input integers. Input: None
	AI Assistant	Solution: We will use the 'math.sqrt' function from the 'math' library to apply the square root to each of the input integers. Here is how to do it: <pre>1 def simplify_fraction(a, b, c, d): 2 sqrt_a = math.sqrt(a) 3 sqrt_b = math.sqrt(b) 4 sqrt_c = math.sqrt(c) 5 sqrt_d = math.sqrt(d)</pre> In this code, 'sqrt_a', 'sqrt_b', 'sqrt_c', and 'sqrt_d' are variables that store the square root of the input integers 'a', 'b', 'c', and 'd' respectively. Next request.
	Human	Abort

Table 18: Application A1, Scenario 1: Autonomous math problem-solving. Preliminary evaluation on MetaGPT. It does not solve the math problem but gives a product solution.

	Action	MetaGPT
	Prompt	Completely simplify and rationalize the denominator: $\frac{\sqrt{160}}{\sqrt{252}} \times \frac{\sqrt{245}}{\sqrt{108}}$
		The boss has tasked me with creating a successful product that is efficient. ## Product Goals <pre> 1 [2 "Create a product that meets the needs of the target market", 3 "Ensure the product is efficient and easy to use", 4 "Achieve a high level of customer satisfaction" 5] </pre> ## User Stories <pre> 1 [2 "As a user, I want a product that is easy to navigate so that I can find 3 what I need quickly", 4 "As a user, I want a product that is reliable and efficient so that I can 5 depend on it", 6 "As a user, I want a product that meets my needs so that I can get value 7 for my money", 8 "As a user, I want a product that is constantly updated and improved so 9 that I can enjoy the best features", 10 "As a user, I want a product that offers good customer support so that I 11 can get help when I need it" 12] </pre> ## Competitive Analysis <pre> 1 [2 "Product A: Offers similar features but lacks efficiency", 3 "Product B: Highly efficient but does not meet all the needs of the 4 target market", 5 "Product C: Has a high level of customer satisfaction but is not easy to 6 navigate", 7 "Product D: Is constantly updated and improved but lacks reliability", 8 "Product E: Offers good customer support but is not efficient", 9 "Product F: Is reliable and efficient but does not offer good customer 10 support", 11 "Our Target Product: Aims to combine all the best features of the 12 competitors while improving on their weaknesses" 13] </pre> ## Competitive Quadrant Chart
	Human	Abort

Table 19: Application A7. Example output of MiniWebChat on one web manipulation task “click-button-sequence.”

	Action	AutoGen
	Init chat	Executor agent to AssistantAgent: We have an autonomous computer control agent that can perform atomic instructions specified by natural language to control computers. There are two types of instructions it can execute. First, given the instruction that matches the regular expression ‘‘~type\s.{1,}\$’’, it can type a list of characters via the keyboard. This instruction should specify the target keyboard input for the agent to type. Before this typing instruction, you should first locate the cursor by clicking the input box with click instruction. Second, given the instruction that matches the regular expression, ‘‘~clickxpath\s.{1,}\$’’, it can click an HTML element with an xpath that is visible on the webpage. The target of this instruction should be a valid xpath. Below is the HTML code of the webpage where the agent should solve a task. <pre> 1 <div id="wrap" data-wob_ref="2" data-wob_eps="e0"> 2 <div id="query">Click button ONE, then click button TWO.</div> 3 <div id="area" data-wob_ref="3" data-wob_eps="e0"> 4 <button id="subbtn" style="position:absolute; left:103px; top:87px" data- wob_ref="4" data-wob_eps="e0">ONE</button> 5 <button id="subbtn2" style="position:absolute; left:44px; top:97px" data- wob_ref="5" data-wob_eps="e0">TWO</button> 6 </div> 7 </div> </pre> task: Click button ONE, then click button TWO. plan: 1. Click the button with xpath “//button[@id='subbtn']”. 2. Click the button with xpath “//button[@id='subbtn2']”. Current task: Click button ONE, then click button TWO. plan:
		***** AssistantAgent to Executor agent: 1. Click the button with xpath “//button[@id='subbtn']”. 2. Click the button with xpath “//button[@id='subbtn2']”. ***** Executor agent to AssistantAgent: Below is the HTML code of the webpage where the agent should solve a task. <pre> 1 <div id="wrap" data-wob_ref="2" data-wob_eps="e0"> 2 <div id="query">Click button ONE, then click button TWO.</div> 3 <div id="area" data-wob_ref="3" data-wob_eps="e0"> 4 <button id="subbtn" style="position:absolute; left:103px; top:87px" data- wob_ref="4" data-wob_eps="e0">ONE</button> 5 <button id="subbtn2" style="position:absolute; left:44px; top:97px" data- wob_ref="5" data-wob_eps="e0">TWO</button> 6 </div> 7 </div> </pre> Current task: Click button ONE, then click button TWO. Here is a plan you are following now. 1. Click the button with xpath “//button[@id='subbtn']”. 2. Click the button with xpath “//button[@id='subbtn2']”. We have a history of instructions that have been already executed by the autonomous agent so far. No instruction has been executed yet. Based on the plan and the history of instructions executed so far, the first instruction should be ‘ ***** AssistantAgent to Executor agent: clickxpath //button[@id='subbtn'] ***** Executor agent to AssistantAgent: Hold on, please wait for my next instruction. ***** AssistantAgent to Executor agent: Sure, I'll be here when you're ready with your next instruction. ***** Executor agent to AssistantAgent: Below is the HTML code of the webpage where the agent should solve a task. <pre> 1 <div id="wrap" data-wob_ref="2" data-wob_eps="e0"> 2 <div id="query">Click button ONE, then click button TWO.</div> 3 <div id="area" data-wob_ref="3" data-wob_eps="e0"> 4 <button id="subbtn" style="position:absolute; left:103px; top:87px" data- wob_ref="4" data-wob_eps="e0" data-tampered="e0">ONE</button> 5 <button id="subbtn2" style="position:absolute; left:44px; top:97px" data- wob_ref="5" data-wob_eps="e0">TWO</button> 6 </div> 7 </div> </pre> Current task: Click button ONE, then click button TWO. Here is a plan you are following now. 1. Click the button with xpath “//button[@id='subbtn']”. 2. Click the button with xpath “//button[@id='subbtn2']”. We have a history of instructions that have been already executed by the autonomous agent so far. 1: clickxpath //button[@id='subbtn'] Based on the plan and the history of instructions executed so far, the next proper instruction should be ‘ ***** AssistantAgent to Executor agent: clickxpath //button[@id='subbtn2'] ***** Executor agent to AssistantAgent: SUCCESS!!!!
		TERMINATE